

PURCHASE REQUEST APPROVAL RECOMMENDER

T49 Mini Project – AI Text Analytics

3–4 Page Assignment Report

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Project Detail	Information
Task	Purchase request approval recommender
Dataset	150 synthetic purchase requests
Approval rules	5-rule approval matrix
AI tool	ChatGPT
Primary outputs	Decision Sheet, Audit, Policy Gap Note, AI Use Log
Final audit sample	30 requests

1. Introduction

The purpose of this mini project was to build an AI-assisted purchase request approval recommender using the supplied synthetic procurement dataset and approval matrix. The system was designed to evaluate each purchase request consistently and recommend one of three outcomes: APPROVE, SEND BACK, or ESCALATE. The recommender also identifies the required approver, missing requirements, and policy triggers so that the recommendation can be reviewed by a human decision-maker.

The project followed the assignment requirements: process all 150 requests, audit 30 decisions, identify policy gaps, document the AI-use process, and provide a final recommendation and reflection. Because the dataset is synthetic, it was suitable for demonstrating the workflow without exposing real personal or confidential procurement information.

2. Objective

The main objective was to translate the five rules in the Approval Matrix into a repeatable AI decision procedure. The system had to use only the supplied request fields and policy rules, avoid inventing missing evidence, and make the reasoning visible. A key design principle was that an AI recommendation is not the same as final authorization: human approvers remain responsible for the actual purchase decision.

- Evaluate all 150 purchase requests against the approval matrix.
- Return a consistent recommendation: APPROVE, SEND BACK, or ESCALATE.
- Identify the applicable rule(s), required approver, and missing evidence.
- Audit a stratified sample of 30 final decisions.
- Use the results to identify weaknesses or ambiguities in the procurement policy.

3. Methodology

The methodology consisted of five stages. First, the workbook was inspected to understand the Purchase Requests and Approval Matrix sheets. Second, the approval rules were converted into a structured prompt. Third, all 150 requests were evaluated and a separate Decision Sheet was created without changing the original data. Fourth, 30 decisions were independently rechecked against the documented rule hierarchy. Finally, the decision results and dataset structure were analysed to identify policy and data gaps.

The supplied matrix contains five core rules: purchases up to ₹50,000 require a department manager and at least one quote; purchases from ₹50,001 to ₹5,00,000 require the department head and Finance with at least three quotes; purchases above ₹5,00,000 require the CFO and Procurement Committee with at least three quotes plus a negotiation note; single-source purchases require CFO approval and written justification; and not-budgeted purchases require CFO approval and a budget re-allocation note.

4. Prompt Design and Iteration

Prompt development was iterative rather than one-shot. The first version focused mainly on purchase-value bands and therefore risked missing special conditions. The second version added single-source and not-budgeted rules, but testing exposed conflicts between quote requirements and the single-source exception. The final version introduced an explicit rule hierarchy, fixed output fields, an evidence-first approach, and a no-invention instruction.

Version	Main issue	Change made	Outcome
V1	Amount bands could miss special rules.	Added checks for single-source and not-budgeted cases.	Rejected.
V2	Single-source and quote rules could conflict; missing documents could be assumed.	Added evidence-first logic and an explicit single-source interpretation.	Improved.
V3	Needed consistent, auditable output.	Fixed output format and required missing evidence to be reported, never invented.	Final version kept.

5. Decision Summary

All 150 requests were processed. The final recommendation distribution shows that most requests could not be recommended for immediate approval because the dataset does not contain some evidence required by the policy.

Decision	Requests	Share	Interpretation
APPROVE	31	20.7%	Mandatory evidence satisfied and no escalation trigger.
SEND BACK	111	74.0%	Missing mandatory evidence or quote/document requirement.
ESCALATE	8	5.3%	CFO/Procurement Committee routing required under the documented hierarchy.
Total	150	100%	All requests evaluated.

The result demonstrates that the recommender is primarily functioning as a completeness and routing checker. In particular, it does not invent negotiation notes or budget re-allocation notes when those fields are absent.

6. Audit and Verification

A stratified audit of 30 decisions was performed: all 8 escalations, 10 approvals, and 12 send-backs. Each sampled recommendation was independently re-applied against the approval matrix and the documented interpretation for special cases. The final audit found 30 matches out of 30, giving 100% final audit agreement.

Audit measure	Result
Requests audited	30
Approvals sampled	10
Send-backs sampled	12
Escalations sampled	8
Independent matches	30 / 30
Final audit agreement	100%

The audit was important because the model's output alone is not sufficient evidence of correctness. The independent re-check confirmed that the final rule hierarchy was being applied consistently. It also highlighted the value of testing edge cases such as single-source purchases and requests marked as not budgeted.

7. Policy Gap Analysis

The analysis identified six important policy or data gaps. The most significant are not model errors; they arise because the current policy requires evidence that the dataset does not capture or because two policy fields are not clearly defined.

Gap	Evidence	Recommended improvement	Priority
Negotiation note missing	52 requests above ₹5,00,000.	Add a negotiation-note field, reference ID and standard template.	High
Budget re-allocation note missing	25 requests are not budgeted.	Add presence, reference and approver fields.	High
Single-source / quote overlap	12 single-source requests are above ₹50,000.	Explicitly state whether single-source overrides normal quote counts.	High
Budget-line flag unclear	40 have approved-budget-line = No; 36 are still marked Budgeted.	Define the meaning and routing effect of the flag.	High
AI approval boundary unclear	All requests still require a named human approver.	Use 'Recommend approval' and require human sign-off.	Medium
No rules for urgency/vendor controls	Urgent, new-vendor and justification details exist but are not governed by the matrix.	Add due-diligence, emergency and justification-quality controls.	Medium

8. Business Insight

The strongest business insight is that a large proportion of requests are being returned because the process cannot prove compliance, not necessarily because the purchases are inappropriate. Of the 150 requests, 52 are above ₹5,00,000 and therefore require a negotiation note, while the dataset has no field for that evidence. Similarly, 25 requests are not budgeted but the workbook has no field for the required budget re-allocation note. The AI therefore correctly exposes a process weakness rather than creating false evidence.

This suggests that the organization would benefit from improving the procurement form itself. Mandatory evidence fields should be captured at the time of request creation so that an AI recommender can validate completeness instead of repeatedly sending requests back for information that should have been collected earlier.

9. AI Use Log and Responsible AI

The workbook contains a completed AI Use Log with more than the required minimum of 15 documented entries. The log records prompt iterations, workbook analysis, processing of the 150 requests, decision summarisation, audit, validation, policy-gap analysis, report drafting and final submission review. Evidence references are retained in the log so that the development process can be traced.

- AI-generated outputs were checked rather than accepted without verification.
- The system was instructed not to invent missing documents or policy evidence.
- The data used for the exercise is synthetic and intended for teaching.
- The final 'APPROVE' outcome means a recommendation, not final authorization.
- Ambiguous policy conditions were documented instead of silently converted into new rules.

10. Recommendation

The organization should implement the recommender as a human-in-the-loop procurement routing and completeness tool, not as an autonomous approval system. Before operational use, the procurement form and policy should be strengthened. In particular, structured fields should be added for negotiation notes and budget re-allocation notes; single-source precedence should be explicitly defined; and the relationship between budget status and the approved-budget-line flag should be clarified.

A second phase should introduce additional controls for new vendors, emergency or urgent procurement, conflict-of-interest checks, vendor due diligence and the quality of purchase justification. These additions would improve the business usefulness of the system without allowing the AI to make unsupported assumptions.

11. Limitations

The dataset is synthetic and the recommender is limited to the information supplied in the workbook. It does not independently assess supplier quality, fraud risk, legal compliance, business necessity, or actual budget availability beyond the available fields. It also cannot verify documents that are not represented in the dataset. Consequently, the tool should be treated as a decision-support and completeness-checking system rather than a substitute for procurement or finance controls.

12. Conclusion and Reflection

The project successfully demonstrated an end-to-end AI-assisted procurement recommendation workflow across 150 purchase requests. The final system produced 31 APPROVE, 111 SEND BACK and 8 ESCALATE recommendations, and the final independent audit achieved 100% agreement on 30 sampled cases. The strongest part of the solution was the evidence-first design: missing information was surfaced rather than guessed.

The main learning from the project is that prompt quality alone cannot solve an incomplete policy. Reliable AI decision support depends on clear rules, structured data, explicit exception handling and independent verification. The project therefore supports a practical recommendation: use AI to improve consistency, routing and evidence checking, while keeping final authorization with accountable human approvers.

End of Report